ModelIt!

Teacher's Guide

Your Body Is a City of Trillions

How Your Organ Systems Work Together to Keep You Alive

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-LS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS1-3

Use argument supported by evidence for how the body is a system of interacting subsystems composed of groups of cells.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how heart rate, blood oxygen, nutrient intake, and physical activity interact as interacting body subsystems.
Disciplinary Core Idea	**LS1.A Structure and Function** — In multicellular organisms, the body is a system of multiple interacting subsystems. These subsystems are groups of cells that work together to form tissues and organs.
Crosscutting Concept	**Systems and System Models** — Students use the concept of interacting systems to understand how changes in one body system (e.g., increased physical activity) cascade through other systems (increased heart rate, breathing rate, energy demand).

CAST Testing Alignment

MULTIPLE CHOICE:

During intense exercise, a student's heart rate increases from 70 bpm to 160 bpm. Which best explains WHY the circulatory system responds this way?

CONSTRUCTED RESPONSE:

A student eats a banana before running a race. Trace the journey of the banana's energy from digestion to muscle contraction, identifying at least three organ systems involved and how they interact.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The human body is an extraordinary system of interacting subsystems — approximately 11 organ systems working in concert to maintain life. Understanding these interactions is fundamental to health science and medicine.

Levels of Organization

The body is organized hierarchically: atoms → molecules → cells → tissues → organs → organ systems → organism. Each level has emergent properties that arise from the interactions of its components. A heart cell can beat on its own, but only the whole heart can pump blood effectively.

The Big Four Systems

The four systems most relevant to exercise physiology are: Circulatory (heart, blood, vessels — transports everything), Respiratory (lungs, airways — gas exchange), Digestive (stomach, intestines — nutrient extraction), and Muscular (muscles — movement and heat production). During exercise, all four must increase their output simultaneously.

Exercise Response

During exercise, muscles demand up to 20x more oxygen and glucose than at rest. The heart rate increases from ~70 to 180+ bpm. Breathing rate jumps from 12 to 40+ breaths/minute. Blood flow is redirected from digestive organs to muscles. These coordinated responses demonstrate how body systems communicate and cooperate.

Homeostasis

Homeostasis is the body's ability to maintain stable internal conditions (temperature, blood sugar, pH, oxygen levels) despite changing external conditions. When you exercise in the heat, your body simultaneously increases heart rate (to feed muscles), produces sweat (to cool down), and adjusts breathing (to maintain oxygen). This multi-system coordination is homeostasis in action.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify heart rate, blood oxygen level, nutrient intake, and physical activity as the key interacting components of the body system.

E **Establish Relationships**
Students determine that physical activity positively drives heart rate (compensation response) while negatively affecting blood oxygen (increased consumption), and that nutrient intake positively supports all body functions.

V **Visualize & Model**
Students build a model showing how the body's subsystems interact to maintain function during different activity levels.

E **Evaluate Outcomes**
Students run resting, sprinting, and low-fuel scenarios to observe how the body compensates — and what happens when it can't.

R **Revise & Extend**
Students add body temperature, lactic acid, or hydration to explore more complex physiological responses.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Right now, your body is running a city of 37 trillion workers — and you're not even thinking about it."*

Watch For

Have students feel their pulse. Count heartbeats for 15 seconds and multiply by 4.

Slide 2: Learning Objectives

3 min

Say: *"Today we're going inside your body to see how four major systems keep you alive."*

Watch For

Have students read objectives. Pre-teach 'homeostasis' and 'organ system.'

Slide 3: Big Question

3 min

Say: *"Your body runs 24/7 for your entire life — no breaks, no software updates, no off switch."*

Watch For

Think-pair-share: What happens if your heart takes a 5-minute break? What about your lungs?

Slide 4: LEVER Framework

2 min

Say: *"We'll model how your body systems talk to each other — especially during exercise."*

Watch For

Preview LEVER steps. Tell students they'll be testing their OWN heart rate data.

Slide 5: Activity 1: Components

8 min

Say: *"What variables does your body need to track? Which can you control and which does your body control automatically?"*

Watch For

Guide sorting. Explain that nutrient intake and activity level are external choices, while heart rate and blood oxygen are internal responses.

Slide 6: Activity 2: Connections

8 min

Say: *"When you sprint, your heart goes UP and your blood oxygen goes DOWN. These are connected!"*

Watch For

Have students do 30 seconds of jumping jacks and immediately retake pulse. Compare to resting rate.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's see what happens when you sprint vs. rest vs. skip lunch!"*

Watch For

Students predict, then run simulations. Compare to their own heart rate data from earlier.

Slide 8: Discoveries

5 min

Say: *"Your body is the most sophisticated system on Earth — every part depends on every other part."*

Watch For

Lead discussion: What would happen to muscles if the circulatory system failed? What about if lungs failed?

Slide 9: STEM Challenge

12 min

Say: *"You're now sports scientists. Design a plan to help a runner perform their best!"*

Watch For

Distribute materials. Groups create evidence-based training and nutrition plans.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Nutrient Intake, Physical Activity
- **INTERNAL:** Heart Rate, Blood Oxygen Level

Relationships

- **Physical Activity to Heart Rate:** POSITIVE (+) — Increased physical activity demands more oxygen and nutrients at muscle cells. The heart responds by beating faster to increase blood flow and delivery.
- **Physical Activity to Blood Oxygen:** NEGATIVE (-) — Intense activity consumes oxygen faster than the lungs can replace it, causing blood oxygen levels to temporarily drop during vigorous exercise.
- **Nutrient Intake to Physical Performance:** POSITIVE (+) — More available nutrients (especially glucose) provide more fuel for cellular respiration in muscle cells, supporting sustained physical activity.

Key Discoveries

- Heart rate increases during exercise to deliver more oxygen and nutrients to working muscles
- Blood oxygen drops during intense activity because muscles consume oxygen faster than lungs can replace it
- Without adequate nutrients, the body can't produce enough energy for sustained physical activity
- All organ systems are interconnected — when one system is stressed, others must compensate

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The heart creates blood"	The heart is a PUMP, not a factory. It pumps blood through blood vessels but doesn't create blood. New blood cells are made in bone marrow. The heart's job is transportation — moving blood where it needs to go, about 2,000 gallons per day.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"You breathe in oxygen and breathe out carbon dioxide"	Air is about 78% nitrogen, 21% oxygen, and 0.04% CO_2. When you exhale, the air is still 78% nitrogen and about 16% oxygen — you only absorb a fraction of the oxygen. You exhale more CO_2 (4%) than you inhale, but the change is smaller than most people think.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Organs work independently"	No organ system works alone. The heart needs signals from the nervous system, oxygen from the respiratory system, and nutrients from the digestive system. Every organ depends on every other organ through the circulatory system's delivery network. Disease in one system cascades to others.